

* MATHEMATICA *

- $D[f[x], \{x, 1\}]$
 $f'[x]$
- $D[f[x], \{x, 2\}]$
 $f''[x]$
- $D[f[x], \{x, n\}]$
 $f^{(n)}[x]$
- $D[f[x, y], \{x, 2\}, \{x, 3\}]$
 $f^{(2, 3)}[x, y]$
- $D[f[x, y, z], \{x, y, z\}]$
 $\{f^{(1, 0, 0)}[x, y, z], f^{(0, 1, 0)}[x, y, z], f^{(0, 0, 1)}[x, y, z]\} \rightarrow \text{gradient}$
- $D[f[u_1[x], u_2[x], x]]$
 $u_2'[x] f^{(0, 1)}[u_1[x], u_2[x]] + u_1'[x] f^{(1, 0)}[u_1[x], u_2[x]]$
- $D[f[\sin[x], \log[x]], \{x, 1\}]$
 $\frac{1}{x} f^{(0, 1)}[\sin[x], \log[x]] + \cos[x] f^{(1, 0)}[\sin[x], \log[x]]$
- $D[x^y, x]$
 x^{-1+y}
- $D[x^y, y]$
 $x^y \log[x]$
- $\partial_x x^y$
 $x^{-1+y} y$
- $\partial_y x^y$
 $x^y \log[x]$